

MATH 448-548

SPRING 2021.

GNU OCTAVE = free MATLAB.

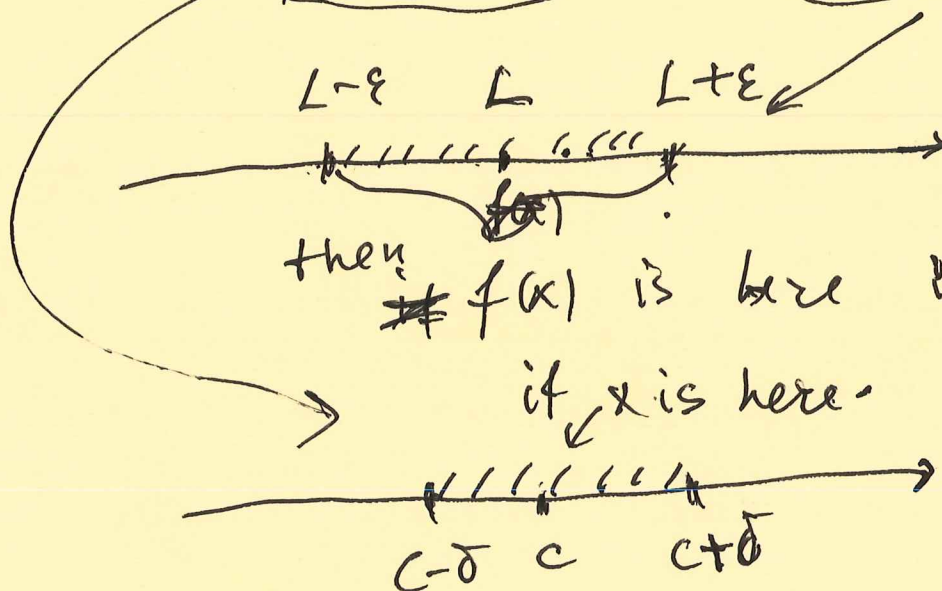
Limit of function

$$\lim_{x \rightarrow c} f(x) = L$$

$$\Uparrow$$

For any $\varepsilon > 0$ ~~$\exists \delta > 0$~~ there is $\delta > 0$
such that

$$0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon$$



2) Continuity $\lim_{x \rightarrow c} f(x) = f(c)$

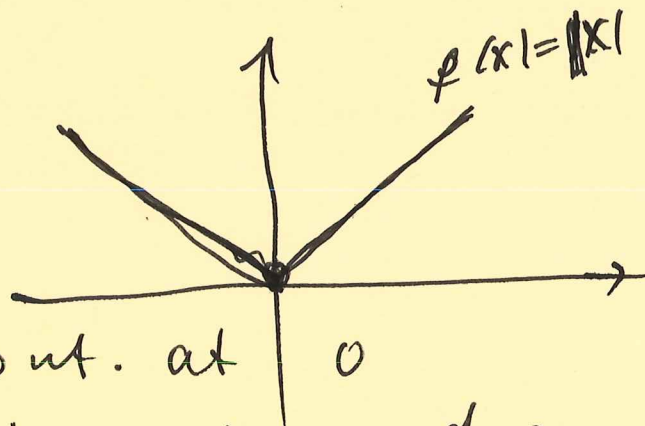
allows to compute limits easily
by plugging in.

3) Derivative

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

$f(x)$ Differentiable at $c \Rightarrow f(x)$ is
continuous at c , but the converse
is generally false

A Continuous function need not be
differentiable.



Ex. $|x|$ is cont. at 0

but the derivative at 0 does not exist

An example produced in the 19th century, gives a function cont. at every point of $\mathbb{R}$, but nowhere differentiable.

Notations:

$C(\mathbb{R})$ - the set of all cont. functions of the real line ($\mathbb{R}$).

$C^k(\mathbb{R})$ - k -times continuously diff. functions of $\mathbb{R}$.

Ex; $C^2(\mathbb{R})$:- $f(x) \in C^2(\mathbb{R})$
means that $f'(x)$ and $f''(x)$ exist
and are continuous functions
at every real x .

(4)

$C^\infty(\mathbb{R})$ - functions with all derivatives of all orders continuous on $\mathbb{R}$.

Ex, 1) $f(x) = x$, is $C^\infty(\mathbb{R})$

$$f'(x) = 1,$$

$$f^{(k)}(x) = f^{(3)}(x) = \dots = f^{(n)}(x) = 0$$

2) $f(x) = x^{1/2}$ - cont. on $\mathbb{R}$
 $[0, \infty)$

$$f'(x) = \frac{1}{2} x^{-1/2} - \text{cont. on } (0, \infty)$$

but not at $x=0$

so $f(x) \in C([0, \infty))$ but

not in $C^1([0, \infty))$

3) $f(x) = |x|$ cont. on $\mathbb{R} \Rightarrow f(x) \in C(\mathbb{R})$

$$f'(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases} \Rightarrow f(x) \notin C^1(\mathbb{R})$$

d.u.e. if $x=0$

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Def. of abs. value:

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Taylor Thm !!!

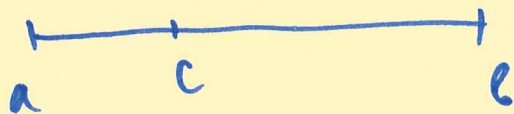
Thm If $f \in C^n([a, b])$ and $f^{(n+1)}$ exists on (a, b) , then for any point c, x in $[a, b]$

$$f(x) = \underbrace{\sum_{k=0}^n \frac{1}{k!} f^{(k)}(c) (x-c)^k}_{T_n(x)} + E_n(x),$$

where $T_n(x)$ - Taylor polynomial of $f(x)$ of degree n

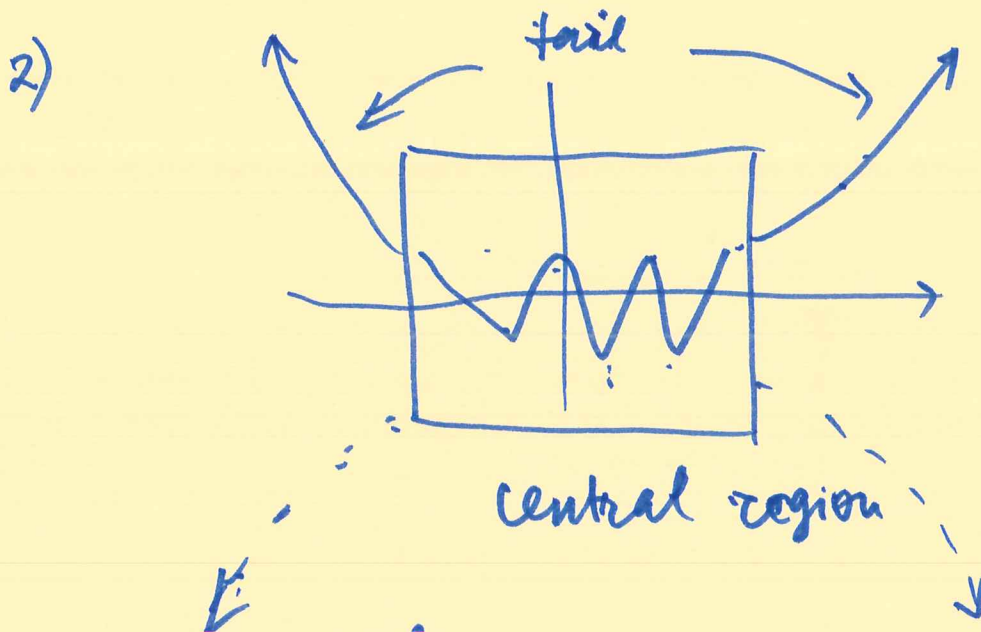
$$E_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi) (x-c)^{n+1}$$

where ξ is a point between c and x



Polynomials - many nice predictable features

1) Polynomials are $C^\infty(\mathbb{R})$ - functions.



3) # of ^{real} zeros $\leq$ the degree

of complex zeros = the degree
(fund. thm. of alg algebra)

Ex. and special cases

1) Let $n = 0$ $f(x) = f(c) + E_0(x)$

$T_n(x) = T_0(x)$ - a constant

function = $f(c)$

$$E_0(x) = \frac{1}{1!} f^{(1)}(c)(x-c)$$

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$$f(x) = f(c) + f^{(1)}(\xi)(x-c), \quad \xi - \text{between } x \text{ and } c.$$

$$f(x) - f(c) = f^{(1)}(\xi)(x-c)$$

Mean Value Thm (MVT).

2) $n=1$ $T_1(x) = \frac{1}{0!} f^{(0)}(c)(x-c)^0 + \frac{1}{1!} f^{(1)}(c)(x-c)^1$

$$= f(c) + f^{(1)}(c)(x-c) - \text{linear function of } x.$$

Taylor thm:

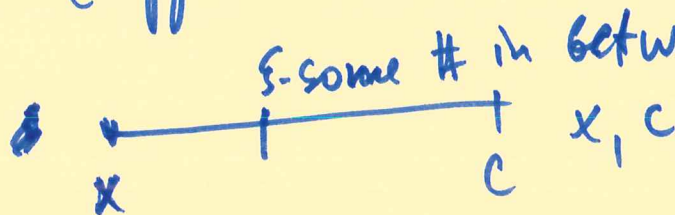
$$f(x) = f(c) + f^{(1)}(c)(x-c) + E_1(x)$$

$$E_1(x) = \frac{1}{2!} f^{(2)}(\xi)(x-c)^2$$

$$f(x) \approx f(c) + f^{(1)}(c)(x-c) - \text{linearization of } f(x) \text{ at } c$$

How do we use $E_1(x)$ -? Normally, derive estimates (upper bound) on

the error.



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Technique: $M = \max_{\xi \text{ is between } x \text{ and } c} |f^{(2)}(\xi)|$

or even $M = \max_{\xi \in [a, b]} |f^{(2)}(\xi)|, \text{ or } \sup_{\xi \in [a, b]} |f^{(2)}(\xi)|$

Then, if $M < \infty$, then

$$|E_1(x)| \leq \frac{1}{2!} M |x-c|^2 = \frac{M}{2} (x-c)^2$$

3) $n=1, f(x) = x^2 + 3x - 2, [a, b] = [0, 1]$

Taylor approximation at $c = \frac{1}{2}$

$$f(x) = T_1(x) + E_1(x)$$

$$T_1(x) = f\left(\frac{1}{2}\right) + f'\left(\frac{1}{2}\right)\left(x - \frac{1}{2}\right) =$$

$$= f\left(\frac{1}{2}\right) + (2x+3)\bigg|_{x=\frac{1}{2}} \left(x - \frac{1}{2}\right) =$$

$$= \left(\frac{1}{4} + \frac{3}{2} - 2\right) + 4\left(x - \frac{1}{2}\right) = -\frac{1}{4} + 4\left(x - \frac{1}{2}\right)$$

$$x^2 + 3x - 2 \approx -\frac{1}{4} + 4\left(x - \frac{1}{2}\right) + E_2(x)$$

$$E_2(x) = \frac{1}{2!} f^{(2)}(\xi) \left(x - \frac{1}{2}\right)^2 =$$

$$= \frac{1}{2!} 2 \left(x - \frac{1}{2}\right)^2 = \left(x - \frac{1}{2}\right)^2$$

(no explicit ξ to worry about).

$$x^2 + 3x - 2 = -\frac{1}{4} + 4\left(x - \frac{1}{2}\right) + \left(x - \frac{1}{2}\right)^2$$

Next, note that for some C^∞ -functions (called real analytic), we can create series expansions (Taylor series)

$$f(x) = \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(c) (x-c)^k$$

(11)

Other formulas for the error
in Taylor thm.

(Thm) if $f \in C^{n+1}([a, b])$, then

for any x, c in $[a, b]$

$$f(x) = T_n(x) + R_n(x)$$

$$R_n(x) = \frac{1}{n!} \int_c^x f^{(n+1)}(t) (x-t)^n dt$$

May be able to use integral inequalities
from analysis to estimate $|R_n(x)|$.

the simplest:

$$|R_n(x)| = \frac{1}{n!} \left| \int_c^x f^{(n+1)}(t) (x-t)^n dt \right| \leq$$

$$\leq \frac{1}{n!} \int_c^x |f^{(n+1)}(t)| |x-t|^n dt$$

$M = \max_{t \in [c, x]} |f^{(n+1)}(t)|$

could be improved. $\int_c^x |x-t|^n dt$
ind of $f(x)$

$$\leq \frac{M}{n!} \int_c^x |x-t|^n dt$$

Multi-dimensional:

$$f(\bar{x}), \quad f(x_1, x_2, \dots, x_m), \quad \bar{x} = (x_1, \dots, x_m)^T$$

Taylor approximation:

$$f(\bar{x}) = f(\bar{c}) + \underbrace{\nabla f(\bar{c}) \cdot (\bar{x} - \bar{c})}_{\text{scalar product}} + \frac{1}{2} \nabla^2 f(\bar{c}) (\bar{x} - \bar{c}) \cdot (\bar{x} - \bar{c}) + E_2(\bar{x})$$

$$\bar{a} \cdot \bar{b} = \sum_{i=1}^m a_i b_i$$

$\nabla^2 f(\bar{c})$ - Hessian matrix ($m \times m$ square

$$(\nabla^2 f)_{jk}(\bar{c}) = \frac{\partial^2 f(\bar{c})}{\partial x_j \partial x_k}$$

symmetric)

$\nabla^2 f$ is the matrix of the 2nd partial derivatives of f .

Orders of Convergence. (1.2 and 2.1....)

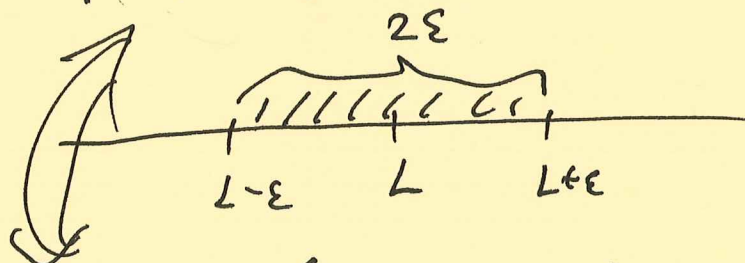
$\{x_n\}$ - a sequence of real #s.

$$\lim_{n \rightarrow \infty} x_n = L$$



for every $\varepsilon > 0$ there is N (pos. integer) such that

$$n > N \Rightarrow |x_n - L| < \varepsilon$$



$$x \in (L - \varepsilon, L + \varepsilon)$$

Ex. $\lim_{n \rightarrow \infty} \frac{n^2 - 1}{n^2 + 1} = 1$

Show this. Pick any $\epsilon > 0$

Consider

$$\left| \frac{n^2-1}{n^2+1} - 1 \right| < \epsilon$$

in terms of

idea: "solve" for n ~~for~~ $\epsilon > 0$ arbitrary

$$-\epsilon < \frac{n^2-1}{n^2+1} - 1 < \epsilon$$

$$-\epsilon < \frac{n^2-1-n^2-1}{n^2+1} < \epsilon$$

$\uparrow$

$$-\epsilon < \frac{1}{n^2+1} < \epsilon$$

$\uparrow$

$$-\frac{2}{\epsilon} < n^2+1 < \frac{2}{\epsilon}$$

$$\left(-\frac{2}{\epsilon} - 1 < n^2 < \frac{2}{\epsilon} - 1 \right)$$

not
informative

$\uparrow$

$$n^2 < \frac{2}{\epsilon} - 1$$

(ϵ small
enough
makes

$$\frac{2}{\epsilon} - 1 > 0$$

$$-\epsilon < \frac{-2}{n^2+1} < \epsilon$$

times -1

$$\epsilon < \frac{2}{n^2+1} < \epsilon$$

$$-\epsilon < \frac{2}{n^2+1} < \epsilon$$

$$n < \sqrt{\frac{2}{\epsilon} - 1}$$

restart

$$-\frac{\epsilon}{2} < \frac{1}{n^2+1} < \frac{\epsilon}{2}$$

not
informative

consider $\frac{1}{n^2+1} < \frac{\epsilon}{2}$

take reciprocals: $n^2+1 > \frac{2}{\epsilon}$

$$n^2 > \frac{2}{\epsilon} - 1$$

$\Downarrow$

$$\sqrt{n^2} > \sqrt{\frac{2}{\epsilon} - 1} \quad (\text{assuming that})$$

since $n > 0$, $\sqrt{n^2} = n$ $\frac{2}{\epsilon} - 1 > 0 \rightarrow$

$$n > \sqrt{\frac{2}{\epsilon} - 1}$$

good for ϵ small
enough)

Point: Given any ϵ (smaller than ^{some} ϵ_0)

find a prescription (algorithm or formula)
for assigning N corresponding to ϵ .

Prescription: take N to be the
smallest positive integer satisfying

$$N > \sqrt{\frac{2}{\varepsilon} - 1}$$

(!!) N is ~~not~~ unique. The prescription
generates a unique $\#$ N ("smallest").

any larger N works as well

(Ex) $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$

$$\begin{aligned} x_{n+1} &= \left(1 + \frac{1}{n+1}\right)^{n+1} \\ x_n &= \left(1 + \frac{1}{n}\right)^n \end{aligned}$$

when we calculate terms, we
find

$$\frac{|x_{n+1} - e|}{|x_n - e|} \rightarrow 1 \quad \text{as } n \rightarrow \infty$$

This suggests thinking about speed
of convergence, not just the fact
of convergence.

$$\boxed{F(x)} \quad x_{n+1} = x_n - \frac{x_n^2}{x_n^2 + x_{n-1}^2} \quad n = 1, 2, 3, \dots$$

$$x_0 = 20.00$$

Here we observe that $\frac{|x_{n+1}|}{|x_n|} \rightarrow 0$

as $n \rightarrow \infty$,

so $|x_{n+1}|$ is "much smaller" than $|x_n| \rightarrow$

$\rightarrow$ a different convergence speed.

Let $\{x_n\}$ be a sequence, $x_n \rightarrow x^*$ as $n \rightarrow \infty$.

Def. 1) the rate of convergence is

at least linear if there is a constant

$C < 1$ and integer $N > 0$ such that

$$(1) \quad |x_{n+1} - x^*| \leq C |x_n - x^*| \quad \text{for all } n \geq N.$$

2) The rate of convergence is at

least superlinear if there is

a sequence $\varepsilon_n \rightarrow 0$ and int. $N > 0$

$$(2) \quad \text{such that } |x_{n+1} - x^*| \leq \varepsilon_n |x_n - x^*| \quad (n \geq N)$$

Convergence of at least order α $\Leftrightarrow$

there is $C > 0$ (not necessarily < 1)
and $\alpha > 0$, and int. $N > 0$ such
that

$$(3) \quad |x_{n+1} - x^*| \leq C |x_n - x^*|^\alpha \quad (n \geq N)$$

$\alpha > 1$ - desirable

$\alpha = 1$ - linear

$\alpha = 2$ - better since $|x_n - x^*| \rightarrow 0$ as

$n \rightarrow \infty \Rightarrow$ will become < 1 eventually

$$\Rightarrow |x_n - x^*|^2 < |x_n - x^*|.$$

Next, consider a framework for
comparing two sequences.

Big O and little o notation.

let $\{x_n\}$, $\{\alpha_n\}$ be two different sequences. We write

$$x_n = O(\alpha_n)$$

$\Leftrightarrow$

$\exists$ (there is) C and n_0 such that

$$|x_n| \leq C |\alpha_n| \quad \forall \text{ (for all) } n \geq n_0$$

The equation $x_n = o(\alpha_n)$ means that

$$|x_n| \leq \varepsilon_n |\alpha_n| \quad \forall n \geq n_0$$

with $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$.

Ex If $x_n \rightarrow 0$ and $\alpha_n \rightarrow 0$, then

$x_n = O(\alpha_n)$ means that two sequences converge "at the same rate"

if $x_n = o(\alpha_n)$ then x_n converges to 0 more rapidly than α_n does.

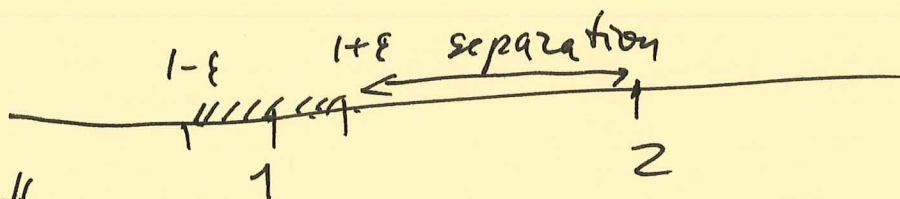
$$\boxed{E_x} \cdot \frac{n+2}{n^2+1} = O\left(\frac{1}{n}\right)$$

$$\frac{1}{n} = O\left(\frac{1}{\ln n}\right)$$

$$\Rightarrow \frac{n(1 + \frac{2}{n})}{n^2(1 + \frac{1}{n^2})} = \frac{1}{n} \quad \left(\frac{1 + \frac{2}{n}}{1 + \frac{1}{n^2}} \right)$$

Since $n \rightarrow \infty$, $\frac{1 + \frac{2}{n}}{1 + \frac{1}{n^2}} \rightarrow 1 \Rightarrow$

$$\frac{1 + \frac{2}{n}}{1 + \frac{1}{n^2}} \leq 2 \quad \text{for all } n \text{ large enough}$$



$$\frac{n+2}{n^2+1} \leq 2 \cdot \frac{1}{n}$$

α_n α_n

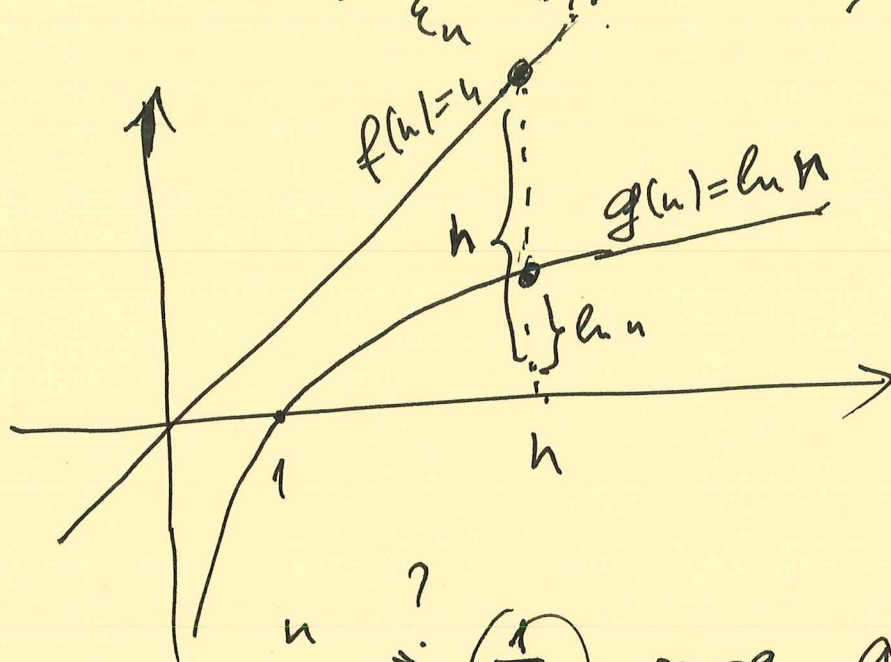
$$\frac{1}{n} = o\left(\frac{1}{\ln n}\right) \quad n \geq 2.$$

little o

Need :

$$\frac{1}{n} \leq \frac{1}{\ln n} \quad \forall n \text{ large enough}$$

$$n \geq \frac{1}{\epsilon_n} \quad n \geq 2.$$



$\frac{n}{\ln n}$ gets larger as $n \rightarrow \infty$

$$\frac{n}{\ln n} \geq \left(\frac{1}{\epsilon_n}\right) \rightarrow \infty \text{ as } n \rightarrow \infty$$

Pointers for getting it work

$$e^{\frac{1}{\epsilon_n} \ln n} \leq e^n$$

$$(e^{\ln n})^{\frac{1}{\epsilon_n}} \leq e^n \Leftrightarrow n^{\frac{1}{\epsilon_n}} \leq e^n$$

$$x_n = O(d_n)$$

$$\Downarrow$$

$$|x_n| \leq C |d_n| \quad \forall n \geq n_0$$

(as $n \rightarrow \infty$)

For functions, this can be adapted.

Ex. $\sin x = x - \frac{x^3}{6} + O(x^5) \quad (\text{as } x \rightarrow 0)$

Meaning: there is an interval $(-c, c)$ (centered at 0) and a constant C such that

$$\left| \sin x - \left(x - \frac{x^3}{6} \right) \right| \leq C |x^5|$$

for all $x \in (-c, c)$

Note: $C < 1$ must to be useful

Justification: Taylor thm. centered at 0

$$\begin{aligned} \sin x &= 0 + \cos 0 \cdot x + \frac{1}{2!}(-\sin 0) x^2 + \frac{1}{3!}(-\cos 0) x^3 + \\ &+ \frac{1}{4!}(\sin 0) x^4 + \frac{1}{5!}(\cos \xi) x^5 \end{aligned}$$

for some ξ between x and 0.

$$\sin x = x - \frac{x^3}{3!} + \frac{1}{5!} \cos \xi x^5$$

$$\sin x - \left(x - \frac{x^3}{3!}\right) = \frac{1}{5!} (\cos \xi) x^5$$

$$\left| \sin x - \left(x - \frac{x^3}{3!}\right) \right| = \frac{1}{5!} |\cos \xi| |x|^5 \leq \\ \leq \frac{1}{5!} 1 \cdot |x|^5$$

$$C = \frac{1}{5!}$$

So, $O(x^5)$ in conjunction with

Taylor formula is used often to

"abbreviate" the error analysis.

Thus,

Implicitly, we often assume that $|f^{(n)}(\xi)|$ is bounded for all ξ in the interval of interval of interest.

(!!) It is ~~dit~~ in general impossible to have that bound, based on mere existence of

$f^{(n)}(x)$. Sufficient conditions for existence of a bound is:

1) interval is closed.

2) $f^{(n)}(x)$ is Continuous on this

interval. Note that continuity of

$f^{(n)}$ is not required for applicability

of the Taylor thm.

(!!) A bound may be possible even if the max value is not attained

Upper and Lower Bounds

supremum — least upper bound

infimum — greatest lower bound

(of a given bounded set)

Bounded set $S \subset \mathbb{R}$ means that

there are $a, b \in \mathbb{R}$ such that

$$a \leq x \leq b \quad \text{for all } x \in S.$$

↓
lower
bound

↑
upper
bound.

Upper and lower bounds are not unique.

Axiom. Any nonempty set of real
#s that has an upper bound, ^{also} has
a least upper bound. (supremum)

Definition. $w = \sup S$ iff

- 1) w is an upper bound for S ;
- 2) if $w < v$ then w is not
an upper bound for S .

$u = \inf S$ iff.

1) u is a lower bound

2) there is ~~not~~ lower bound larger than u for S .

Q: is $\sup S$ an element of S ?

Ex., let $S = \{x: x^2 < 2\}$

Find $\sup S$.

$$x^2 < 2$$

$$\Updownarrow$$

$$|x| < \sqrt{2}$$

$$\Updownarrow$$

$$-\sqrt{2} < x < \sqrt{2}$$

$$\sqrt{x^2} = |x|$$

$\sqrt{2}$ is an upper bound, but not an element of S . is there a smaller upper bound.

like $\sqrt{2} - \varepsilon$, $\varepsilon > 0$ (small). Negative.

Since if $x = \sqrt{2} - \varepsilon$ then

$$y = \sqrt{2} - \frac{\varepsilon}{2} \text{ satisfies } y > x, y \in S$$

so x is not an upper bound.

Conclusion $\sup S = \sqrt{2}$, but $\sqrt{2} \notin S$.

(!!) $\sup S$ is not $\max S$ in general

$\max S$ is an element of S by def.

$\sup S$ is not.

$\Downarrow$

$$\max S \leq \sup S$$

if $\max$ exists.

in the above example, $\max S$ does not exist.

$$\forall x \in S \quad \exists \varepsilon > 0 : \quad x = \sqrt{2} - \varepsilon < \sqrt{2}$$

$$\text{and } y \in S = \sqrt{2} - \frac{\varepsilon}{2} > x$$

$$y < \sqrt{2}$$

(!!!) It is much safer to write $\sup S$ and not $\max S$.

Computer Arithmetic

Storing $\#$ s in memory requires truncation
every $\#$ stored in a fixed $\#$ of bits

round off error appears as a consequence.
on a typical laptop - desktop $\sim 10^{-14}$

"truncation noise" $\Rightarrow$ always present

Let

x be an actual real $\#$

$fl(x)$ - its machine stored approximation

More notations: if x^* is approximated

by x , then

absolute error: $|x - x^*|$

relative error: $\frac{|x - x^*|}{|x|}$

Ex (Round-off error contamination)

if $x \approx y$ then approximating

$x - y$ with $fl(x) - fl(y)$.

may result in $fl(x) - fl(y) = 0$, and

then the relative error is

$$\frac{|x - y - (fl(x) - fl(y))|}{|x - y|} = 1 \quad (100\%)$$

more generally if $|fl(x) - fl(y)| \ll |x - y|$

(much smaller), then the relative error may be close to 1.

Loss of significance.

(Ex.) Avoiding loss of significance by transforming the expression

~~Ex~~ $y = x - \sin x$, $x \approx 0$

for small x , $\sin x \approx x \Rightarrow$

write

danger!

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

//

$$x - \sin x \approx \frac{x^3}{3!} - \frac{x^5}{5!} + \frac{x^7}{7!} \approx y(x)$$

Stable and unstable computations.

Conditioning.

1. Instability $\rightarrow$ amplification of small error in a recursive computation

Ex. $x_0 = 1, \quad x_1 = \frac{1}{3}$

$$x_{n+1} = \frac{13}{3} x_n - \frac{4}{3} x_{n-1}$$

This computes $x_n = \left(\frac{1}{3}\right)^n$

Check this: let x_{n-1} be given, $x_{n-1} = \left(\frac{1}{3}\right)^{n-1}$

then

$$x_{n-2} = \left(\frac{1}{3}\right)^{n-2}$$

$$x_n = \frac{13}{3} x_{n-1} - \frac{4}{3} x_{n-2} =$$

$$= \frac{13}{3} \left(\frac{1}{3}\right)^{n-1} - \frac{4}{3} \left(\frac{1}{3}\right)^{n-2} = \left(\frac{1}{3}\right)^{n-2} \left(\frac{13}{3} \cdot \frac{1}{3} - \frac{4}{3} \right) =$$

$$= \left(\frac{1}{3}\right)^{n-2} \left(\frac{13-12}{9} \right) = \left(\frac{1}{3}\right)^{n-2} \frac{1}{9} = \left(\frac{1}{3}\right)^{n-2+2} = \left(\frac{1}{3}\right)^n.$$

This algorithm is unstable $\rightarrow$ see book.

The relative error of computing

x_{15} is about 10^8 .

Reason: error in x_n is multiplied by

$\frac{13}{3} > 1$ each time.

Conditioning.

Focus on linear equations.

$A \bar{x} = \bar{b}$ - a linear system.

$$\bar{x} \in (x_1, \dots, x_n)^T, \quad \bar{b} = (b_1, \dots, b_n)^T$$

$A_{n \times n}$ - a matrix

Condition # of A is

$$\kappa(A) = \|A\| \|A^{-1}\|$$

Small κ - ok. large κ - not good

[Ex]. $A = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$

$$A \bar{x} = (\lambda_1 x_1, \lambda_2 x_2, \dots, \lambda_n x_n)^T$$

The system:

$$\lambda_i x_i = b_i \quad i = 1, \dots, n.$$

$$x_i = \frac{b_i}{\lambda_i} \quad \left(\begin{array}{l} \text{assume that} \\ \lambda_i \neq 0 \text{ so} \\ A \text{ is invertible} \end{array} \right)$$

Suppose $\lambda_1 = 1$, and $\lambda_n = 10^{-9}$

so let $\|A\| = \max_i |\lambda_i| = 1$

Then $A^{-1} = \begin{pmatrix} \frac{1}{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & \frac{1}{\lambda_n} \end{pmatrix} \Rightarrow \|A^{-1}\| = \frac{1}{\lambda_n} =$

$$= 10^9$$

$$\kappa(A) = 1 \cdot 10^9 = 10^9$$

Significance: a small error in b_n

say δ amplifies to $\frac{\delta}{\lambda_n} = 10^9 \delta$

so x_n is way off.

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Solutions of non-linear equations, (Ch. 3)

Bisection Method.

(Thm) (Intermediate Value)

On an interval $[a, b]$, a cont.
function assumes all values between
 $f(a)$ and $f(b)$.

Useful implication: if

$$f(a)f(b) < 0 \quad (\text{opposite signs}) \quad (1)$$

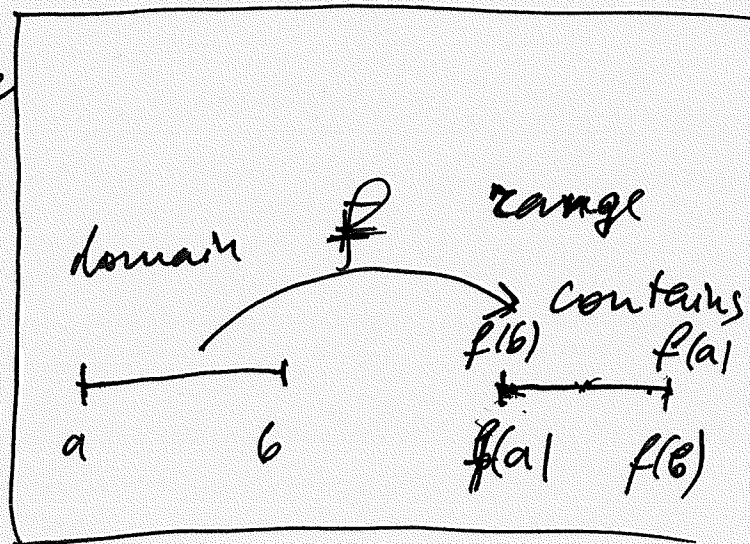
then $\exists c \in [a, b]$ such that $f(c) = 0$.

This ^{ensures} ~~helps to~~ existence
of solutions to
equations

$$f(x) = 0 \quad (*)$$

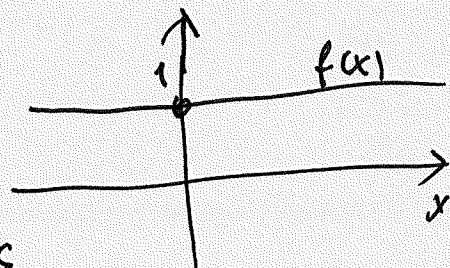
for $x \in [a, b]$

and f continuous,



Existence is not guaranteed

Ex. 1) $f(x) = 1$.



$1 = 0$ - no solutions

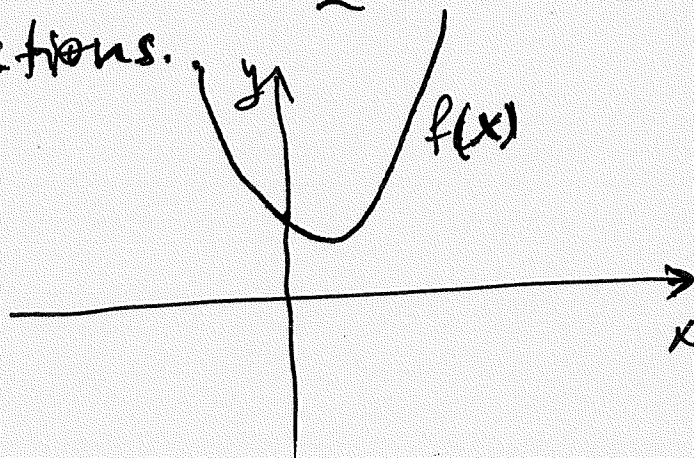
2) $f(x) = x^2 + x + 1$

try solving

$$x^2 + x + 1 = 0$$

$$x = \frac{-1 \pm \sqrt{1 - 4}}{2} = \frac{-1 \pm \sqrt{-3}}{2}$$

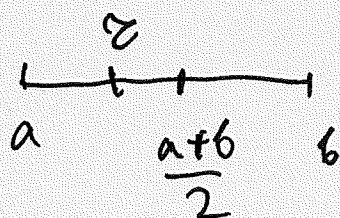
no real solutions.



Bisection method

1. Find (by trial and error) $[a, b]$ such that $f(a)f(b) < 0$. (if possible).

2. By IVT, there is $c \in (a, b)$ such that $f(c) = 0$.



Goal: to pinpoint ϵ more precisely
 so, try evaluating $f\left(\frac{a+b}{2}\right)$. If
 $f\left(\frac{a+b}{2}\right) = 0$, we are done, $c = \frac{a+b}{2}$ is
 a solution.

If not then either $f(a)f(c) < 0$ or
 $f(b)f(c) < 0$ (or ~~to~~ both)

If $f(a)f(c) < 0$, set $b := c$ and
 repeat.

If $f(a)f(c) \geq 0$, ~~set~~ then $f(b)f(c) < 0$

set $a := c$ and repeat

Stop if $|f(c)| < \epsilon$ (ϵ - a given
 tolerance)

Error Analysis.

Nested intervals:

$$\begin{array}{ccccccc} \text{length } l & & l/2 & & l/4 & & l/8 \\ [a_0, b_0] & \supset & [a_1, b_1] & \supset & [a_2, b_2] & \supset & [a_3, b_4] \dots \end{array}$$

$$a_0 \leq a_1 \leq a_2 \leq \dots \leq b_0$$

$$b_0 \geq b_1 \geq b_2 \geq \dots \geq a_0$$

$$\text{also, } b_{n+1} - a_{n+1} = \frac{1}{2} (b_n - a_n) = \frac{1}{2} \cdot \frac{1}{2} (b_{n-1} - a_{n-1}) = \dots$$

Writing it out we obtain

$$b_n - a_n = 2^{-n} (b_0 - a_0)$$

$$\Downarrow$$

$$\lim_{n \rightarrow \infty} (b_n - a_n) = 0$$

Thus, if $\lim b_n$ and $\lim a_n$ both exist, then

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} a_n = c.$$

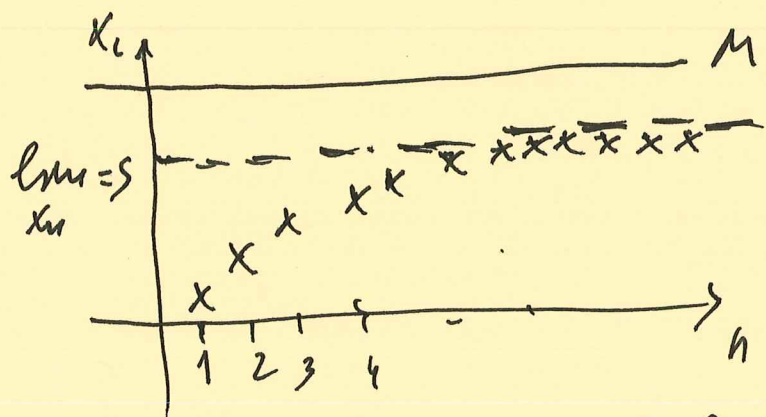
Existence of the limit follow from
the ~~gene~~ analysis thm. on monotonic
sequence.

Thm if $\{x_n\}_{n=0}^{\infty}$ is increasing and
bounded above, then

$$\lim_{n \rightarrow \infty} x_n = S \quad \text{where } |S| < \infty$$

if $\{x_n\}$ is decreasing and bounded
below, then $\lim_{n \rightarrow \infty} x_n = S$; $|S| < \infty$

$$|x_n| < M$$



!!! then $\lim x_n = \sup \{x_n\}$ if $x_n \nearrow$
 $\lim x_n = \inf \{x_n\}$ if $x_n \searrow$.

(39)

Explanation for $\lim a_n = \lim b_n = r$

We know that r is some $\#$. Show that $f(r) = 0$.

$$0 \geq f(a_n) f(b_n) \quad (2)$$

Pass to the limit $n \rightarrow \infty$ on both
since f is continuous,

$$\lim f(a_n) = f(\lim a_n) = f(r)$$

$$\lim f(b_n) = f(\lim b_n) = f(r)$$

Thus, from (2) $\Rightarrow$

$$0 \geq f(r) f(r) = [f(r)]^2$$

$\Downarrow$

$$f(r) = 0.$$

Convergence speed? Let $c_n = \frac{a_n + b_n}{2}$

$$|r - c_n| \leq \frac{1}{2} |b_n - a_n| = 2^{-(n+1)} (b_0 - a_0)$$

(!!) Overall, the method converges, but the speed may be slow.

Inf.

Many solutions on a finite interval?

(40)

possible

Next, consider the Newton's Method.

Newton's method converges faster than bisection method (when it [↑] does)

~~Quadr.~~ @ Quadratic rate for N. method
vs,

at linear most for the bisection method.

let r be such that $f(r) = 0$.

Taylor approximation: $x+h=r$

$$0 = f(r) = f(x+h) \begin{matrix} \downarrow \text{centered} \\ \text{at } x \end{matrix} \quad (x = r-h)$$

$$f(x) + f'(x)h + O(h^2) \approx$$

$$\approx f(x) + f'(x)h$$

(*) - small

$$\downarrow \\ x \approx r$$

Solve for h :

$$0 \approx f(x) + f'(x)h$$

$$h \approx - \frac{f(x)}{f'(x)}$$

(would be nice
to have $f'(x) \neq 0$)

This motivates

Newton's method

$$\boxed{x_{n+1} = \underbrace{x_n}_{\text{like } x} - \underbrace{\frac{f(x_n)}{f'(x_n)}}_{\text{like } h}}$$

compare with

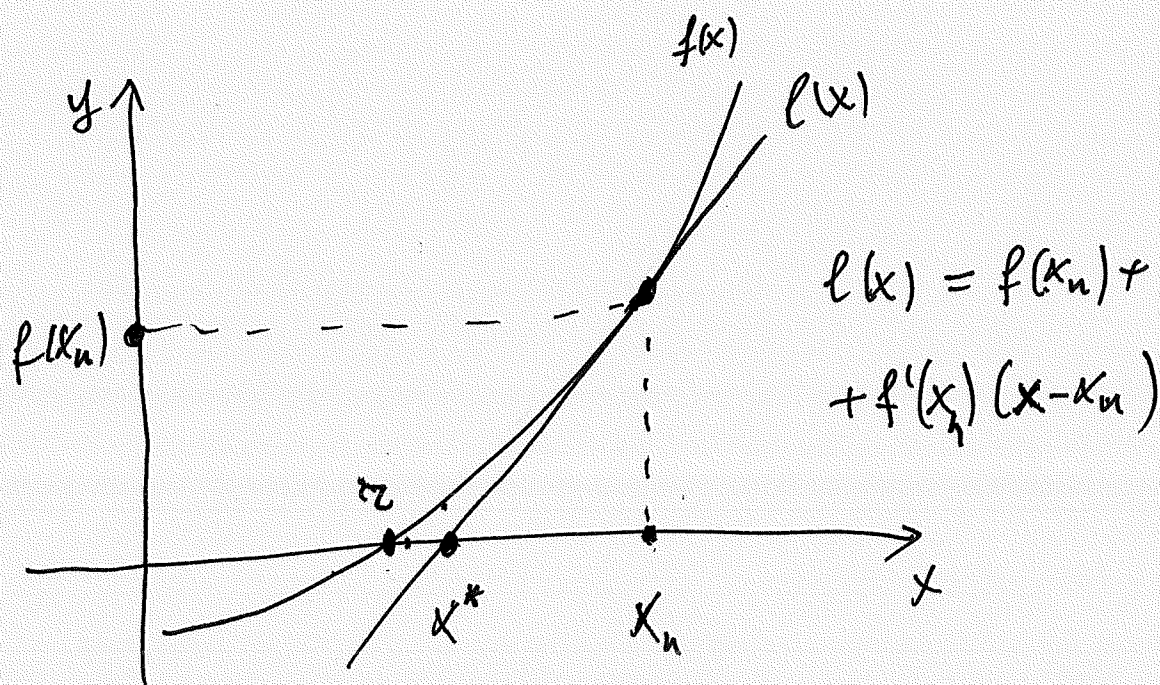
$$z = x + h$$

Start with some guess x_0

$$\text{then } x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

etc.



$$l(x^*) = 0 \Leftrightarrow f(x_n) + f'(x_n)(x^* - x_n) = 0$$

$$x^* - x_n = - \frac{f(x_n)}{f'(x_n)}$$

$$x^* = x_n - \frac{f(x_n)}{f'(x_n)} \Rightarrow x^* = x_{n+1}$$

So this is the x_{n+1} in the Newton's method. Looks nice! (?)

Error Analysis for Newton's Method.

(43)

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$\{x_n\}$ approximates ~~the~~^a solution τ of

$f(\tau) = 0$, starting from the initial

guess x_0 .

Error $e_n = x_n - \tau$

Assume that :

(i) f'' is cont.

(ii) $f(\tau) = 0$, $f'(\tau) \neq 0$. (τ is a simple zero)

Consider

$$e_{n+1} = x_{n+1} - \tau = x_n - \frac{f(x_n)}{f'(x_n)} - \tau =$$

$$= e_n - \frac{f(x_n)}{f'(x_n)} = \frac{e_n f'(x_n) - f(x_n)}{f'(x_n)} \quad (1)$$

$$0 = f(r) = f(x_n - e_n) \stackrel{\text{Taylor}}{=} \underbrace{\hspace{10em}}_{\text{near}}$$

centered

at x_n keep two terms +
error

$$= f(x_n) - e_n f'(x_n) + \frac{1}{2} e_n^2 f''(\xi_n)$$

for some ξ_n between x_n and r $\Downarrow$

$$e_n f'(x_n) - f(x_n) = \frac{1}{2} f''(\xi_n) e_n^2 \quad (2)$$

Plug (2) into (1):

$$e_{n+1} = \frac{1}{2} \frac{f''(\xi_n)}{f'(x_n)} e_n^2 \stackrel{\text{preliminary}}{\sim} \frac{1}{2} \frac{f''(r)}{f'(x_n)} e_n^2$$

$$\sim \left(\frac{1}{2} \frac{f''(r)}{f'(r)} \right) e_n^2 := C e_n^2 \rightarrow \text{may expect}$$

quadratic rate of convergence.